Numerical study of coupled Ginzburg-Landau Equations

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Introduction

1 Modern overview

2 Targets and goals

Pulse propagation in erbium-doped fiber (EDF) can be described by the following coupled Ginzburg-Landau equations:

$$\begin{cases} \frac{\partial u}{\partial z} = \frac{i\Delta\beta}{2}u - \delta\frac{\partial u}{\partial T} - \frac{i\beta_2}{2}\frac{\partial^2 u}{\partial T^2} + \frac{g}{2}u + \frac{g}{2\Omega_g^2}\frac{\partial^2 u}{\partial T^2} + i\gamma\left(|u|^2 + \frac{2}{3}|v|^2\right)u + \frac{i\gamma}{3}v^2u^*, \\ \frac{\partial v}{\partial z} = \frac{i\Delta\beta}{2}v - \delta\frac{\partial v}{\partial T} - \frac{i\beta_2}{2}\frac{\partial^2 v}{\partial T^2} + \frac{g}{2}v + \frac{g}{2\Omega_g^2}\frac{\partial^2 v}{\partial T^2} + i\gamma\left(|v|^2 + \frac{2}{3}|u|^2\right)u + \frac{i\gamma}{3}u^2v^*, \end{cases} \tag{1}$$

where u and v are slowly varying pulse envelopes. Our goal is to study different effects of pulse propagation using numerical simulations of Eq. 1.

Chapter 1

Analytical analysis

1 Particular analytical solutions of original equation

Let us consider first equation of Eq. 1, assuming $v = 0$:

$$\frac{\partial u}{\partial z} = \frac{i\Delta\beta}{2}u - \delta\frac{\partial u}{\partial T} - \frac{i\beta_2}{2}\frac{\partial^2 u}{\partial T^2} + \frac{g}{2}u + \frac{g}{2\Omega_g^2}\frac{\partial^2 u}{\partial T^2} + i\gamma\left|u\right|^2 u. \tag{1.1}$$

Using traveling wave reduction in the form:

$$u\left(z,T\right) = y(\mathrm{w})e^{i(kz-\omega T)}, \quad \text{where} \quad \mathrm{w} = z - c_0 T, \tag{1.2}$$

and evaluating corresponding derivatives:

$$\begin{cases} \frac{\partial u}{\partial z} = \left(y_{\mathrm{w}} + iky\right)e^{i(kz-\omega T)}, \\ \frac{\partial u}{\partial T} = \left(-c_0 y_{\mathrm{w}} - i\omega y\right)e^{i(kz-\omega T)}, \\ \frac{\partial^2 u}{\partial T^2} = \left(c_0^2 y_{\mathrm{ww}} + 2i\omega c_0 y_{\mathrm{w}} - \omega^2 y\right)e^{i(kz-\omega T)}, \end{cases} \tag{1.3}$$

we can rewrite Eq. 1.1 as:

$$\begin{aligned} &\left[\frac{c_0^2 g}{2\Omega_g^2} - i\left(\frac{c_0^2\beta_2}{2}\right)\right]\cdot y_{\mathrm{ww}} + \left[\delta c_0 - 1 + \beta_2\omega c_0 + i\left(\frac{c_0 g\omega}{\Omega_g^2}\right)\right]\cdot y_{\mathrm{w}} + \\ &+ \left[\frac{g}{2}\left(1 - \frac{\omega^2}{\Omega_g^2}\right) + i\left(\frac{\Delta\beta}{2} + \delta\omega + \frac{\beta_2}{2}\omega^2 - k\right)\right]\cdot y + i\gamma y^3 = 0. \end{aligned} \tag{1.4}$$

Extracting real and imaginary parts of Eq. 1.4, we obtain over-defined system of equations in the form:

$$\begin{cases} \left(\frac{c_0^2 g}{2\Omega_g^2}\right)\cdot y_{\mathrm{ww}} + \left(\delta c_0 - 1 + \beta_2\omega c_0\right)\cdot y_{\mathrm{w}} + \left[\frac{g}{2}\left(1 - \frac{\omega^2}{\Omega_g^2}\right)\right]\cdot y = 0, \\ \left(-\frac{\beta_2 c_0^2}{2}\right)\cdot y_{\mathrm{ww}} + \left(\frac{c_0 g\omega}{\Omega_g^2}\right)\cdot y_{\mathrm{w}} + \left(\frac{\Delta\beta}{2} + \delta\omega + \frac{\beta_2\omega^2}{2} - k\right)\cdot y + \gamma\cdot y^3 = 0. \end{cases} \tag{1.5}$$

At parameters:

$$\begin{cases} \delta = \frac{1-\beta_2 c_0 \omega}{c_0}, \\ g = 0, \end{cases} \tag{1.6}$$

the equation for real part (second equation from system 1.5) can be satisfied, while imaginary part equation can be written as:

$$-\left(\frac{\beta_2 c_0^2}{2}\right) y_{\mathrm{ww}} + \left(-\frac{\beta_2 \omega^2}{2} + \frac{\omega}{c_0} - k + \frac{\Delta\beta}{2}\right) y + \gamma y^3 = 0 \tag{1.7}$$

Eq. 1.7 is a second-order nonlinear autonomous differential equation. It takes the form:

$$y_{\mathrm{ww}} + \xi_1 y + \xi_2 y^3 = 0, \tag{1.8}$$

where

$$\begin{cases} \xi_1 = -\frac{\omega^2}{c_0^2} + \frac{2\omega}{c_0^3 \beta_2} - \frac{2k}{c_0^2 \beta_2} + \frac{\Delta\beta}{c_0^2 \beta_2}, \\ \xi_2 = \frac{2\gamma}{c_0^2 \beta_2}. \end{cases} \tag{1.9}$$

After multiplying Eq. 1.9 on y_{w} and integrating, we obtain first integral:

$$\frac{y_{\mathrm{w}}^2}{2} + \frac{\xi_1}{2} y^2 + \frac{\xi_2}{4} y^4 = I_1. \tag{1.10}$$

The first derivative of wave envelope then takes form:

$$y_{\mathrm{w}} = \pm \sqrt{2I_1 - \xi_1 y^2 - \frac{\xi_2}{2} y^4}, \tag{1.11}$$

or

$$\pm \left(\mathrm{w} - \mathrm{w}_0\right) = \int \frac{dy}{\sqrt{2I_1 - \xi_1 y^2 - \frac{\xi_2}{2} y^4}}. \tag{1.12}$$

Integral 1.12 can be expressed via Jacobi Elliptic Functions, and cannot be computed analytically. By introducing new variable $y\left(\mathrm{w}\right) = \sqrt{Y\left(\mathrm{w}\right)}$, Eq. 1.10 can be written with lower nonlinearity degree as follows:

$$Y_{\mathrm{w}}^2 + 2\xi_2 Y^3 + 4\xi_1 Y^2 - 8I_1 Y = 0. \tag{1.13}$$

Next, we apply the Simplest Equations Method [cite] to Eq. 1.13. We will be looking for $Y\left(\mathrm{w}\right)$ in form:

$$Y\left(\mathrm{w}\right) = M_0 + M_1 Q\left(\mathrm{w}\right) + M_2 Q^2\left(\mathrm{w}\right), \tag{1.14}$$

where $Q\left(\mathrm{w}\right)$ is the solution of Riccati equation:

$$Q_{\mathrm{w}} = \mu\left(Q^2 - Q\right), \tag{1.15}$$

of the form:

$$Q(\mathrm{w}) = \frac{1}{1 + \exp\left(\mu(\mathrm{w} - \mathrm{w}_0)\right)}. \tag{1.16}$$

Using 1.14 and 1.15, 1.13 can be written as:

$$< bigpolynom > \tag{1.17}$$

2 Numerical methods description

Chapter 2

Numerical Analysis

Chapter 3

Conclusion